

POOJA KUMARI GUPTA

PhD., Physics

Department of Physics, Indian Institute of Technology Guwahati
Theoretical Quantum Physics Researcher

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Research Profile

I am a research scholar working in theoretical quantum physics, with a focus on hybrid quantum systems and cavity magnonics.

Research Interests

Hybrid quantum systems	Cavity magnonics
Cavity magnomechanics	Cavity optomagnonics
Quantum nonlinear optics	Driven-dissipative systems
Quantum correlations	Nonlinear dynamics

Education

Ph.D. in Physics <i>Department of Physics, Indian Institute of Technology Guwahati</i> Research area: Hybrid Quantum Systems and Cavity Magnonics	2021–Present
M.Sc. in Physics <i>University / National Institute of Technology, Rourkela</i> CGPA: 7.84 / 10	2017–2019
B.Sc. in Physics <i>University / St. Xavier's College, Ranchi</i> Percentage: 81.56	2014–2017
Senior Secondary <i>School / Delhi Public School, Ranchi</i> Percentage: 80.10	2011–2013
High School <i>School / Saraswati Vidya Mandir, Gumla</i> CGPA: 9.4	2001–2011

Research Experience

Research Scholar <i>Department of Physics, Indian Institute of Technology Guwahati</i>	2021–Present
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Publications

- Quantum transduction of a superconducting qubit in an electro-optomechanical and an electro-optomagnonical system**, Roson Nongthombam, Pooja Kumari Gupta, Amarendra K Sarma, *Physical Review A*, 108, 043501 (2023).
- Quantum-interference-induced magnon blockade and antibunching in a hybrid quantum system**, Pooja Kumari Gupta, Sampreet Kalita, Amarendra K Sarma *Journal of the Optical Society of America B*, 41, 447-455 (2024).
- Quantum signatures of bistability and limit cycle in Kerr-modified cavity magnomechanics** Pooja Kumari Gupta, Subhadeep Chakraborty, Sampreet Kalita, Amarendra K Sarma, *Physical Review A*, 110, 063504 (2024).

Conference Participation

Photonics 2024 IIT Kharagpur, India Oral Presentation: "Quantum Correlations and Bistability in a Kerr modified Cavity Magnomechanical System"	December 2024
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Awards and Fellowships

- **Ph.D. Fellowship**, MHRD, Indian Institute of Technology Guwahati, 2021–Present.
- **Award**, Best oral presentation award, photonics 2024.

Technical Skills

Programming	Python, QUTIP, Matlab
Symbolic Computing	Mathematica
Scientific Writing	LaTeX, Overleaf

Teaching Experience

Institute/Organization	Role	Courses	Year
IIT Guwahati, India	Teaching Assistant	PH110	2022–2023
IIT Guwahati, India	Teaching Assistant	PH101	2024–2025
NPTEL	Teaching Assistant	Quantum Entanglement: Fundamentals, Measures and Applications	2024–2025
NPTEL	Teaching Assistant	Essentials of Quantum Optics	2025
NPTEL	Teaching Assistant	Quantum Technology and Quantum Phenomena in Macroscopic Systems	2025

Position of Responsibility

- Vice president, SPIE IITG Student Chapter (2022-2024)

References

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